

July 6, 2026 04:13 AM GMT

Semiconductor Capital Equipment | North America

Rolling over PTs to 2028

\$200bn of WFE in 2027 is well understood, and \$250bn in 2028 is starting to be priced in. We see opportunity in MKS and ONTO, where 2027 earnings power remains underappreciated. We are slightly less enthusiastic on the large caps.

What's Changed? We revise up our 2026 forecast from \$149bn (+27% y/y) to \$155bn (+32%), our 2027 forecast comes up from \$191bn (+28%) to \$202bn (+31%) and our 2028 forecast comes up from \$215bn (+13%) to \$227bn (+13%). The majority of our 2026-27 revisions are from our intra-quarter memory WFE revisions, and our 2028 revisions are driven by foundry logic.

\$200bn in 2027 is priced in. Our large-cap US SPE coverage now trades at 37x our CY27 forecasts, which have been adjusted to reflect our updated WFE outlook. That 37x multiple represents a 75% premium to the 21x through-cycle average at which these stocks have traded since 2020, and a 20%+ premium to the most recent peak of 30x in July 2024. We think \$200bn of WFE in 2027 is well understood by the market and are therefore shifting our valuation base year to 2028. Based on our \$227bn WFE forecast for 2028, our large-cap coverage now trades at 33x, implying that **\$250bn of WFE in 2028 may already be close to priced in.**

\$250bn+ in 2028 is possible, but we are not ready to underwrite it. Over the past two weeks, 2028 WFE expectations have increased significantly, but we question both the legitimacy of pricing in such upside during 2026 and the feasibility of a \$250bn WFE environment. We are not doubting the possibility of \$250bn+, but the key questions we are asking ourselves are: **1) Memory:** Can the market digest \$80bn+/\$30bn+ of DRAM/NAND WFE, which would imply bit supply growth north of 40%? **2) EUV:** Can ASML supply the approximately 130 tools required? **3) Logic:** Are we willing to underwrite a meaningful contribution (\$10bn+) from the likes of TeraFab and Rapidus? We are continuing to work through these questions, but without greater comfort, we are not yet ready to reflect \$250bn+ in our models.

We revise up target multiples for KLA and AMAT, as we see the broadening of leading-edge customers benefiting KLA, while the strength of DRAM WFE in CY26 should benefit AMAT. Given that we have yet to see NAND WFE revisions, we leave target multiples for LAM and MKS unchanged. We revise down multiples for NVMI and CAMT, as we have questions around whether NVMI can outgrow WFE in 2027 and have market share concerns in HBM for CAMT (see [Exhibit 3](#) for details).

Across our US SPE coverage, our preference is for companies whose earnings power on \$200bn of WFE in 2027 remains underappreciated. Specifically, we think Top Pick MKS and OW-rated ONTO present substantial upside from undemanding valuations on 2027 earnings. We remain OW on KLA and LAM, but our enthusiasm toward both names is slightly more muted, as we think expectations for \$250bn+ of WFE in 2027 are already partially priced in.

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SEMICONDUCTOR CAPITAL EQUIPMENT		
North America		
Industry View	In-Line	
WHAT'S CHANGED		
Camtek (CAMT.O)	From	To
Price Target	\$163.00	\$167.00
KLA Corp (KLAC.O)	From	To
Price Target	\$190.00	\$274.00
Lam Research Corp (LRCX.O)	From	To
Price Target	\$331.00	\$404.00
MKS Inc. (MKSI.O)	From	To
Price Target	\$374.00	\$442.00
Nova Ltd (NVMI.O)	From	To
Price Target	\$494.00	\$540.00
Applied Materials Inc. (AMAT.O)	From	To
Price Target	\$502.00	\$647.00

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For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

WFE Forecast Table

Exhibit 1:

(\$mn)	2021	2022	2023	2024	2025	2026e	2027e	2028e
WFE Revenue	91,092	96,222	96,008	105,057	117,048	154,638	201,887	227,269
Semi Revenue	553,540	573,978	526,820	630,549	791,061	1,607,055	1,959,883	1,959,883
Semi Capex	157,788	187,982	176,924	198,540	215,972	305,309	357,665	351,202
Y/Y Change %								
WFE Revenue	43%	6%	0%	9%	11%	32%	31%	13%
Semi Revenue	26%	4%	-8%	20%	25%	103%	22%	0%
Semi Capex	33%	19%	-6%	12%	9%	41%	17%	-2%
Metrics								
WFE Intensity	16%	17%	18%	17%	15%	10%	10%	12%
WFE% of Total Capex	58%	51%	54%	53%	54%	51%	56%	65%
Semi Capital Intensity	29%	33%	34%	31%	27%	19%	18%	18%
3rd Party Sources								
Semi	87,499	94,100	95,610	104,270	115,700	126,100	135,200	
Gartner	92,843	101,101	102,820	111,777	124,452	144,886	161,086	154,674
WFE by segment								
Foundry/Logic	50,265	59,180	70,023	69,192	75,307	89,479	114,491	133,729
Memory	39,828	35,854	24,794	35,228	41,118	64,261	86,438	92,215
DRAM	20,664	18,256	19,579	29,244	31,083	48,621	62,430	64,108
NAND	19,164	17,598	5,215	5,985	10,035	15,640	24,009	28,107
Other	999	1,188	1,192	637	622	898	957	1,324
Y/Y Change %								
Foundry/Logic	45%	18%	18%	-1%	9%	19%	28%	17%
Memory	42%	-10%	-31%	42%	17%	56%	35%	7%
DRAM	53%	-12%	7%	49%	6%	56%	28%	3%
NAND	31%	-8%	-70%	15%	68%	56%	54%	17%
Other	19%	19%	0%	-47%	-2%	44%	7%	38%
% of Total								
Foundry/Logic	55%	62%	73%	66%	64%	58%	57%	59%
Memory	44%	37%	26%	34%	35%	42%	43%	41%
DRAM	23%	19%	20%	28%	27%	31%	31%	28%
NAND	21%	18%	5%	6%	9%	10%	12%	12%
Other	1%	1%	1%	1%	1%	1%	0%	1%
WFE Intensity								
Semi ex-Memory revenue	399,706	444,153	434,511	465,032	568,550	727,417	835,151	835,151
Memory revenue	153,834	129,825	92,309	165,516	222,512	879,638	1,124,732	1,124,732
DRAM	92,960	77,769	51,945	94,860	150,598	581,901	675,059	675,059
NAND	55,953	47,109	36,104	66,427	67,685	293,508	445,443	445,443
WFE Intensity								
Foundry Logic	13%	13%	16%	15%	13%	12%	14%	16%
Memory	26%	28%	27%	21%	18%	7%	8%	8%
DRAM	22%	23%	38%	31%	21%	8%	9%	9%
NAND	34%	37%	14%	9%	15%	5%	5%	6%
% of Memory WFE								
DRAM	52%	51%	79%	83%	76%	76%	72%	70%
NAND	48%	49%	21%	17%	24%	24%	28%	30%

Source: Gartner, SEMI, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 2:

(\$mn)	2021	2022	2023	2024	2025	2026e	2027e	2028e
WFE by Region								
Total	91,092	96,222	96,008	105,057	117,048	154,638	201,887	227,269
North America	7,755	10,239	11,346	13,657	10,534	12,680	16,555	18,636
Europe	2,879	5,982	5,741	4,202	2,341	2,629	3,028	3,409
Japan	6,879	7,244	6,438	7,354	8,779	10,825	14,536	16,591
Korea	22,757	19,655	18,965	18,910	23,410	35,567	48,453	52,954
Taiwan	23,566	26,566	19,501	15,233	27,506	40,515	56,528	63,635
China	24,695	22,105	31,682	42,548	41,552	48,711	57,941	68,181
Other	2,561	4,431	2,334	3,152	2,926	3,711	4,845	3,864
WFE by Region (Y/Y)								
North America	28%	32%	11%	20%	-23%	20%	31%	13%
Europe	8%	108%	-4%	-27%	-44%	12%	15%	13%
Japan	14%	5%	-11%	14%	19%	23%	34%	14%
Korea	51%	-14%	-4%	0%	24%	52%	36%	9%
Taiwan	54%	13%	-27%	-22%	81%	47%	40%	13%
China	48%	-10%	43%	34%	-2%	17%	19%	18%
Other	40%	73%	-47%	35%	-7%	27%	31%	-20%
WFE by Region (% of Total)								
North America	9%	11%	12%	13%	9%	8%	8%	8%
Europe	3%	6%	6%	4%	2%	2%	2%	2%
Japan	8%	8%	7%	7%	8%	7%	7%	7%
Korea	25%	20%	20%	18%	20%	23%	24%	23%
Taiwan	26%	28%	20%	15%	24%	26%	28%	28%
China	27%	23%	33%	41%	36%	32%	29%	30%
Other	3%	5%	2%	3%	3%	2%	2%	2%

Source: Gartner, SEMI, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 3: Multiple revision rationale

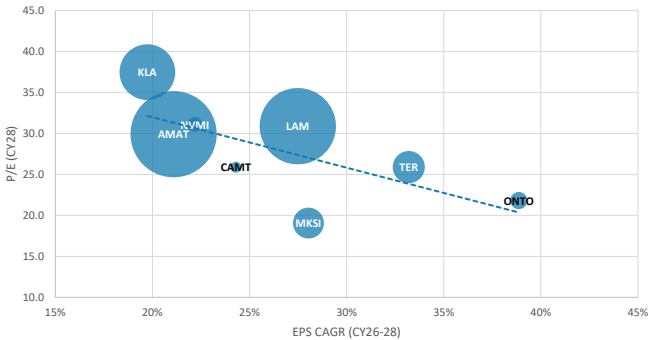
	Old PT	New PT	Old Multiple (CY27)	New Multiple (CY28)	Multiple revision rationale
KLA	\$190	\$274	33x	38x	Broadening of leading edge logic customers (Intel, Rapidus, TeraFab)
AMAT	\$502	\$647	28x	30x	DRAM WFE far stronger than anticipated in CY26
LAM	\$331	\$404	34x	34x	Waiting for NAND WFE revisions to reward multiple expansion
MKS	\$374	\$442	22x	22x	
NVMI	\$494	\$540	36x	33x	Questions around outgrowth vs WFE in 2027
CAMT	\$163	\$167	32x	27x	Market share worries in HBM vs ONTO/KLA

Source: Morgan Stanley Research

How do we think about SPE valuation?

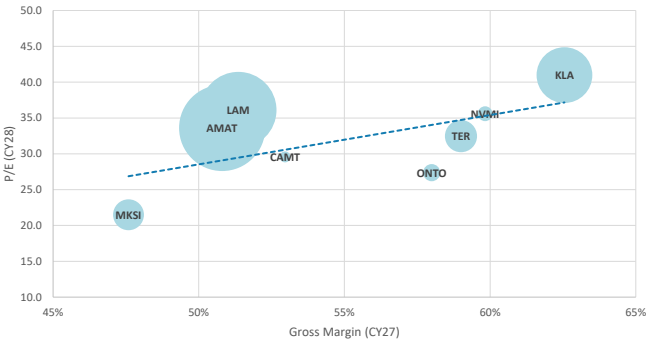
Our guiding principle for US SPE coverage valuation is that multiples are closer aligned to gross margin rather than growth rates. Gross margin is reflective of the “value” equipment companies provide to customers, which in the case of process tools is throughput while for inspection it would be yield. Gross margin for SPE doesn’t materially vary y/y given these companies aren’t rewarded/punished for where we are in the cycle. Gross margin expansion comes from operational improvement or providing customers with more value, rather than opportunistically increasing pricing.

Exhibit 4: US SPE CY28 P/E vs CY26-28 EPS CAGR



Source: FactSet, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 5: US SPE CY28 P/E vs CY27 Gross Margin (%)



Source: FactSet, Morgan Stanley Research. e = Morgan Stanley Research estimates

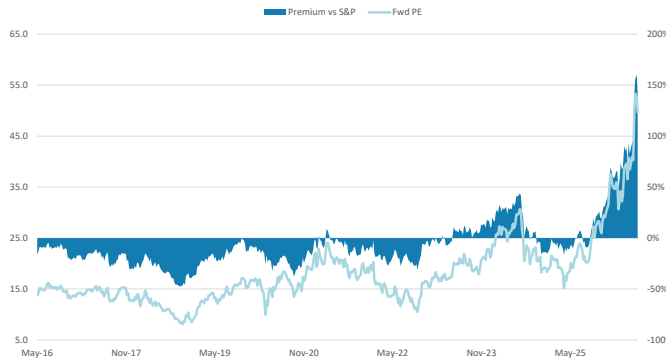
Our large cap US SPE coverage now trades at 37x CY27, a 20%+ premium to the most recent peak of 30x in Jul. 2024 when the market was increasingly looking towards \$125bn of WFE in 2025. Based on our \$227bn WFE in 2028, our large caps now trade at 33x, implying that \$250bn in 2028 may be close to being priced in.

Exhibit 6: AMAT/LAM/KLA FwdPE vs 2020- average



Source: FactSet, Morgan Stanley Research. Note: average since Jan 2020

Exhibit 7: AMAT/LAM/KLA FwdPE & Premium vs S&P



Source: FactSet, Morgan Stanley Research.

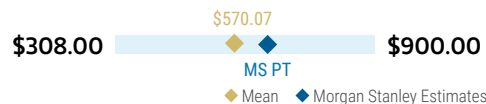
Risk Reward – Applied Materials Inc. (AMAT.O)

DRAM/Leading Logic upside & China/ICAPS derisked

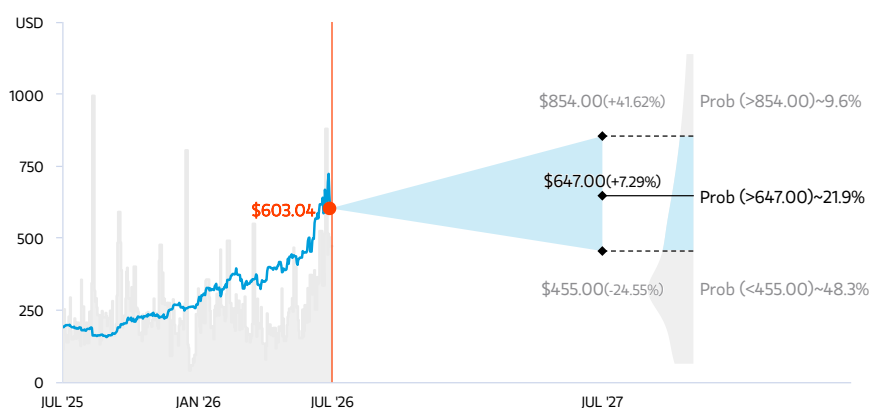
PRICE TARGET \$647.00

~30x CY28e EPS of \$21.55, a 4-turn discount to LAM and 8-turns to KLA to reflect growth prospects in DRAM but concerns around market share loss in foundry logic.

Consensus Price Target Distribution



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



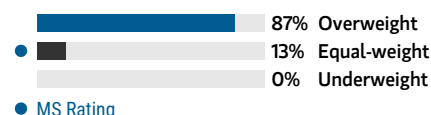
Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 2 Jul 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

EQUAL-WEIGHT THESIS

■ AMAT has traded at a material valuation discount vs LAM & KLA given execution and concerns around market share loss in China. We think AMAT will be a share gainer in 2026 but given our growth forecast is in-line with broader WFE in 2027 we think the valuation discount is unlikely to narrow in the near term.

■ AMAT has the highest DRAM mix in their portfolio (CY26 31% vs LAM 23%, KLA 28%) and is most levered to greenfield wafer additions. We see DRAM as the fastest growing end-market in 2026 but the slowest growing in 2027.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$854.00

~35x CY28 EPS of \$24.41

DRAM & ICAPS remain strong and AMAT's margin expansion accelerates.

- Revenue grows 24.6% in CY28 as AMAT outperforms WFE
- Gross margin expands to 51.9% in CY27
- Multiples expand to 35x PE, as equipment is perceived as a growth sector within semis

BASE CASE

\$647.00

~30x CY27 EPS of \$21.55

AMAT grows in-line with WFE with DRAM growth slightly offset by China/ICAPS

- Revenue growth of 13.3% in CY28
- Non-GAAP gross margin slightly increases to 51.4% in CY27
- Multiple at a premium to AMAT's historical average

BEAR CASE

\$455.00

~24x CY28 EPS of \$18.96

DRAM WFE slows in 2027 and leading edge logic sees headwinds from foundries scaling back their aspirations.

- Revenue increases 4.2% in CY28 as WFE recovery is subdued.
- Gross margin slightly increases to 50.6% in CY27
- Multiple contracts to a 24x PE, 2 standard deviation premium to historical levels

Risk Reward – Applied Materials Inc. (AMAT.O)

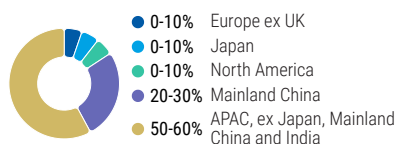
KEY EARNINGS INPUTS

Drivers	Oct 2025	Oct 2026e	Oct 2027e	Oct 2028e
GAAP Revenue (\$, mm)	28,368	33,656	44,233	51,269
MW Gross Margin (%)	48.8	50.0	51.0	51.4
Non-GAAP EPS (\$)	9.42	12.38	17.41	20.97
Inventory (\$, mm)	5,915	7,717	9,156	10,201
DOI	146.2	164.9	151.8	147.2

INVESTMENT DRIVERS

- DRAM WFE outpaces overall WFE growth
- AMAT capitalizes on logic growth opportunities (GAA, back-side power)

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	3/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Market share gains as DRAM outpaces WFE
- AMAT captures market share in logic architecture inflections (backside power)

RISKS TO DOWNSIDE

- Widespread restrictions on equipment shipments and services to China
- AMAT loses market share with key customers (Samsung, TSMC)

OWNERSHIP POSITIONING

Inst. Owners, % Active	50.7%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	29.5%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Oct 2026e

Sales / Revenue (\$, mm)	32,921		34,104
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EBITDA (\$, mm)	9,014		12,935
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Net income (\$, mm)	9,411		10,832
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EPS (\$)	11.73		12.70
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◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Camtek (CAMT.O)

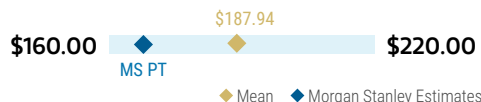
HBM beneficiary, but fairly priced

PRICE TARGET \$167.00

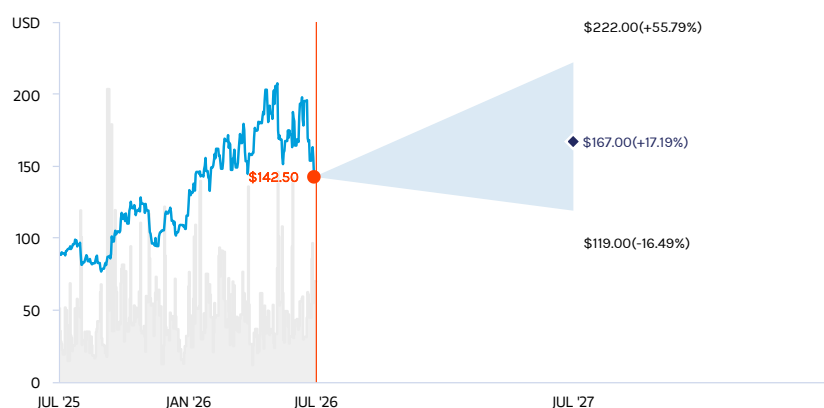
28x multiple on the company's CY28 EPS of \$5.51. Multiple is a 5-turns premium to the company's through cycle average and a 10-20% discount to peer NOVA.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

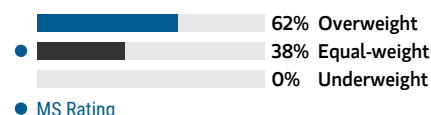
Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

■ We like CAMT's structural growth story led by metrology/inspection intensity increasing in the back-end and the CAMT specific growth story driven by HBM. We think CAMT is set up for double-digit growth through 2026-27 led by the drivers above and continued OSAT capex strength.

■ CAMT trades at above through-cycle average multiples on our above-consensus estimates, which keeps us EW the name. We would be inclined to turn more positive if we get indication that CAMT can achieve our bull case EPS of \$5.50+ in 2027 but stay sidelined for now.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$222.00

~33x multiple on CY28 EPS of \$6.32 and net cash

We reflect stronger than expected OSAT capex for 2026-2027 as well as market share gain and a multiple closer to early cycle.

- Revenue growth of 25.3% in CY28
- GM expands to 54.7% and EPS to \$6.32 in CY27

BASE CASE

\$167.00

~28x multiple on CY28 EPS of \$5.51 and net cash

Multiple is roughly in-line with the company's through cycle average.

- Revenue growth of 13.9% in CY28
- CY28 GM of 53.5%
- EPS of \$5.51 in CY28

BEAR CASE

\$119.00

~22x multiple on CY28 EPS of \$4.81 and net cash

We reflect weaker than expected OSAT capex for 2026-2027 as well as market share loss and a multiple closer to late cycle.

- Revenue growth of 5.9% in CY28
- GM declines to 52.0% and EPS comes in at \$4.81 in CY28

Risk Reward – Camtek (CAMT.O)

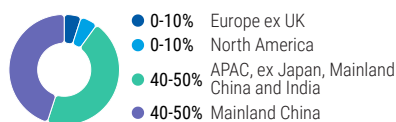
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
GAAP Revenue (\$, mm)	496	577	731	833
MW Gross Margin (%)	51.0	51.8	53.0	53.0
Non-GAAP EPS (\$)	3.19	3.56	4.82	5.51
Inventory (\$, mm)	112	124	137	151
DOI	164.4	159.2	142.3	138.2

INVESTMENT DRIVERS

- HBM capacity expansion
- OSAT capex strength
- Market share vs ONTO & KLA

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

RISKS TO PT/RATING

RISKS TO UPSIDE

- OSAT capex growth accelerates
- HBM capacity growth accelerates
- Share gain from ONTO and KLA

RISKS TO DOWNSIDE

- OSAT capex growth decelerates due to end-market weakness
- HBM capacity growth slower than expected
- Share loss to ONTO and KLA

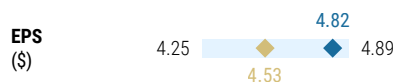
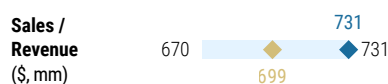
OWNERSHIP POSITIONING

Inst. Owners, % Active	84.7%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	29.5%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – KLA Corp (KLAC.O)

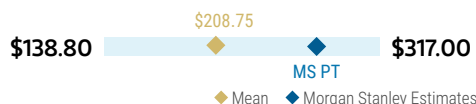
OW on structural and idiosyncratic drivers

PRICE TARGET \$274.00

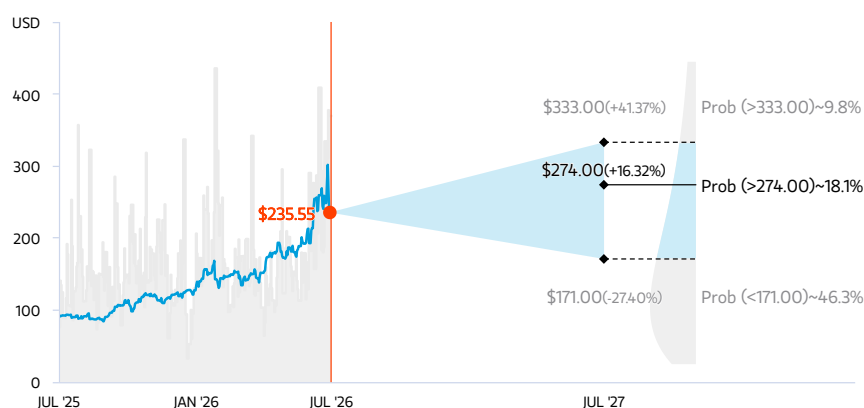
38x CY28e EPS of \$7.22, a 15% premium to US SPE peers LAM & AMAT to reflect growth prospects in foundry logic and DRAM.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



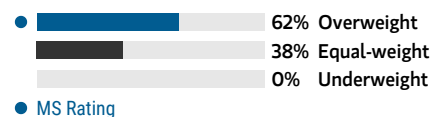
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 2 JUL 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

OVERWEIGHT THESIS

■ KLA's fundamental story has gone from strength to strength: 1) Process control intensity: to structurally increase going forward due to larger die sizes and increased design starts, both causes of lower yield. 2) Share increase: KLA should continue to gain market share in process control given the company is first to inflections and has the fundamental technology to enable that. ■ We believe a premium multiple to SPE peers is justified given KLA's growth prospects (outperformance vs WFE) as well as its superior gross margin.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*
Pricing Power: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$333.00

41x CY28e EPS of \$8.12

KLA gains more market share than expected on the back of leading edge logic spending accelerating and process control intensity in DRAM increasing greater than expected.

- In this scenario we assume a considerable premium to US SPE peers (AMAT/LAM) as KLA's growth prospects are more appreciated.

- Revenue grows 27.7% in CY28 as KLAC sees outsized growth due to increased market share.

- GM expands to 63.5% and EPS grows to \$8.12 in 2028.

BASE CASE

\$274.00

38x CY28e EPS of \$7.22

KLA grows ahead of WFE in 2027 due to increased process control intensity in leading edge logic and DRAM.

- In this scenario we assume a premium to US SPE peers (AMAT/LAM) that is slightly higher than historic due to KLA's growth prospects.

- Revenue grows 16.1% in CY28 and KLA slightly increases market share.

- GM expands to 62.8% and EPS grows to \$7.22 in 2028

BEAR CASE

\$171.00

26x CY28e EPS of \$6.59

KLA's market share contracts due to leading edge logic and DRAM investment slowing down. KLA faces increasing competition in mask inspection, pressuring valuation multiple.

- In this scenario we assume multiple in-line with US SPE peers (AMAT/LAM)

- Revenue growth of 7.3% in CY28 as leading edge logic growth is limited.

- GM declines to 61.8% and EPS slightly increases to \$6.59 in 2028

Risk Reward – KLA Corp (KLAC.O)

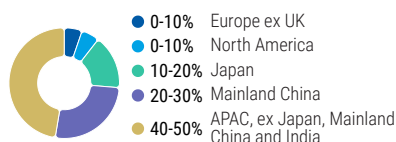
KEY EARNINGS INPUTS

Drivers	Jun 2025	Jun 2026e	Jun 2027e	Jun 2028e
GAAP Revenue (\$, mm)	12,156	13,524	18,117	21,746
MW Gross Margin (%)	62.3	62.2	62.4	62.7
MW EPS (\$)	3.33	3.70	5.41	6.66
Inventory (\$, mm)	3,212	3,623	4,918	5,586
DOI	246.7	251.5	258.1	246.8

INVESTMENT DRIVERS

- WFE market growth, specifically in leading edge logic where KLA has outsized exposure.
- Process control intensity and KLA's market share within process control.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	3/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Foundries (TSMC, Intel, Samsung) pull forward their logic roadmaps
- Design starts in leading edge nodes increase
- Process control intensity increases in DRAM due to HBM

RISKS TO DOWNSIDE

- Certain foundries (Intel, Samsung) scale back their aspirations
- Market share loss in mask inspection to Lasertec or in e-beam to AMAT or ASML
- China restrictions

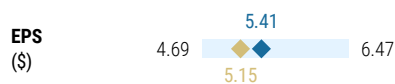
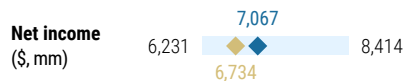
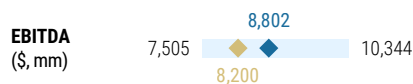
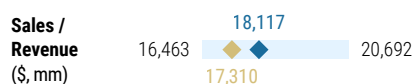
OWNERSHIP POSITIONING

Inst. Owners, % Active	51.2%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	29.5%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Jun 2027e



Source: Refinitiv, Morgan Stanley Research

Risk Reward – Lam Research Corp (LRCX.O)

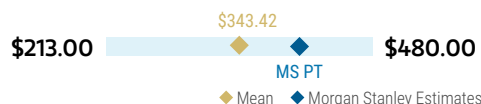
WFE outperformance intact with NAND & Logic

PRICE TARGET \$404.00

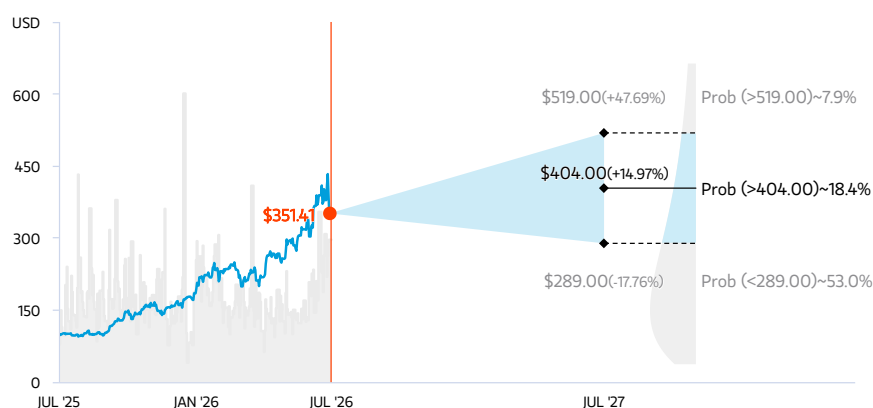
34x CY28 non-GAAP EPS of \$11.89, multiple reflects a 15% premium to AMAT (30x) to reflect LAM's market share gains.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



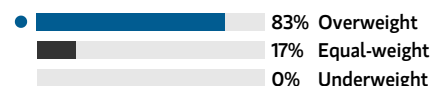
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 2 Jul 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

OVERWEIGHT THESIS

- After shipments underperformed WFE by 33ppt in 2023 due to a cyclical decline in NAND and China entity list impact, LAM outperformed WFE by 8ppt in 2024 and we estimate another 27ppt outperformance in 2025 driven by NAND, China, and TSMC. We model LAM's system shipments to outperform WFE in 2026 & 2027 for the 3rd and 4th consecutive year as NAND and non-China foundry logic continues to strengthen.
- LAM has unique drivers supporting the long-term growth story as the company is seeing gross margin expansion from shifting manufacturing to Malaysia, is positioned well to gain share with the 4F2 transition in DRAM and in advanced packaging.

Consensus Rating Distribution



● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$519.00

~38x CY28 non-GAAP EPS \$13.66

DRAM and NAND pricing and utilization improve further than expected, and LAM sees record revenue in CY27. The company sees revenue growth exceed WFE on the back of market share gains in leading edge nodes.

- Revenue growth of 13.2% in CY28 as LAM benefits from NAND WFE growth and market share gains

- GM expands to 52.0% in CY28

BASE CASE

\$404.00

~34x CY28 non-GAAP EPS of \$11.89

We expect LAM to outperform in 2026-27 on the back of NAND and foundry logic strength.

– Revenue growth of 13.2% in CY28 driven by NAND and foundry logic

- CY28 GM of 52.0% as regional mix deterioration is offset by top-line growth

BEAR CASE

\$289.00

~27x CY28 non-GAAP EPS of \$10.72

Memory markets turn weaker and LAM loses share to competitors.

- Revenue growth of 25.6% in CY28 as memory recovery more muted than expected

- GM of 53.0% in CY28

Risk Reward – Lam Research Corp (LRCX.O)

KEY EARNINGS INPUTS

Drivers	Jun 2025	Jun 2026e	Jun 2027e	Jun 2028e
GAAP Revenue (\$, mm)	18,436	23,223	33,063	40,245
MW Gross Margin (%)	48.8	50.2	51.0	51.8
Non-GAAP EPS (\$)	4.13	5.73	8.78	11.22
Inventory (\$, mm)	4,308	4,583	5,945	6,391
DOI	164.0	142.3	132.0	118.5

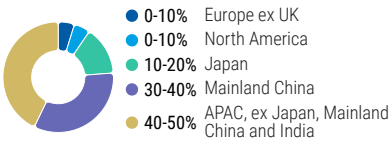
CATALYST CALENDAR

Date	Event	Source: Refinitiv, Morgan Stanley
04 Nov 2026 - 08 Nov 2026	Lam Research Corp Annual Shareholders Meeting	

INVESTMENT DRIVERS

- NAND WFE market trends (LAM has outsized share)
- Market share gains from new logic architectures (GAA, backside power)

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	2/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Market share gains in new logic architectures (GAA, backside power)
- NAND capex recovers to new highs
- Continued share gains in China

RISKS TO DOWNSIDE

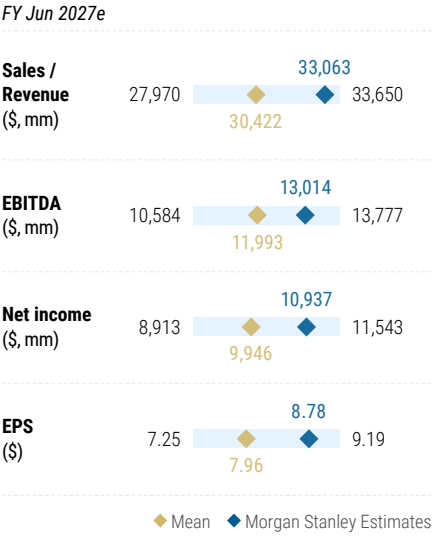
- NAND WFE weaker than expected
- Widespread export restrictions to China
- Market share loss to competitors in etch

OWNERSHIP POSITIONING

Inst. Owners, % Active	48.6%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	29.5%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS



Source: Refinitiv, Morgan Stanley Research

Risk Reward – MKS Inc. (MKSIO) Top Pick

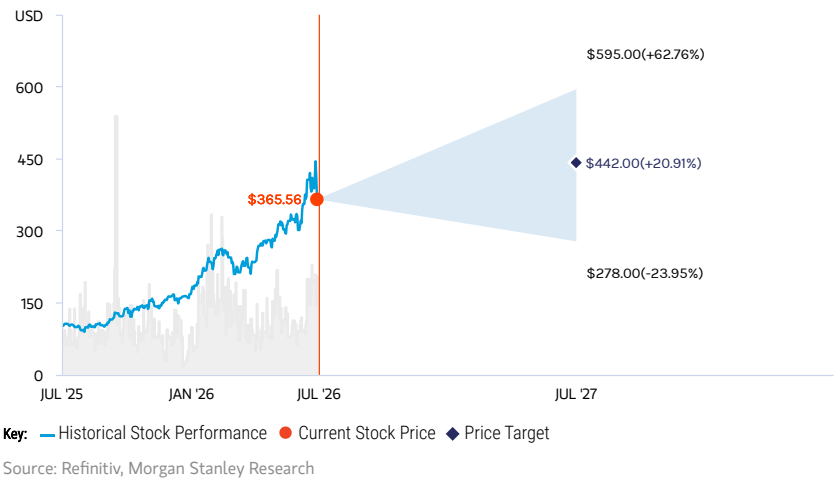
Cyclical recovery to drive deleverage & EPS growth, OW

PRICE TARGET **\$442.00**

22x FY28 EPS of \$20.08. Multiple is a larger discount to front-end SPE vs historic to reflect balance sheet concerns and business mix shift to packaging.



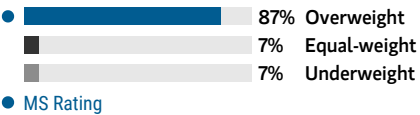
RISK REWARD CHART



OVERWEIGHT THESIS

- Cyclical recoveries for MKS' semi and electronics & packaging (E&P) business should drive revenue and FCF growth, helping the company delever and EPS outperform.
- The company's outsized NAND exposure has worked against the company during 2023-2024, but we expect NAND spend to outgrow overall WFE in 2027 and MKS to be a beneficiary.
- We forecast MKS' E&P business to grow above PCBs but below advanced electronics through 2027, as the company's "Optimize the Interconnect" strategy drives share gains from scale.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*
Self-help: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	\$595.00	BASE CASE	\$442.00	BEAR CASE	\$278.00
25x FY27 EPS of \$23.79		22x FY28 EPS of \$20.08		16x FY28 EPS of \$17.38	
A stronger than expected WFE and advanced electronics upcycle drives more aggressive debt paydowns and MKS' EPS sees strong growth.		MKS' semiconductor & E&P businesses benefit from a cyclical recovery in 2027 and grows revenue y/y in both segments.		A weaker WFE and advanced electronics cycle dampen the recovery and MKS sees slower debt repayments and deleverage.	
-Revenue grows 17.6% in 2028 on the back of a stronger WFE and advanced electronics upcycle.		-Revenue grows 9.9% in 2028.		-Revenue growth of 2.2% in 2028 on the back of a weaker WFE and advanced electronics upcycle.	
-GPM% expands to 48.9% and EPS to \$23.79 in 2028.		-GPM% expands to 47.9% and EPS to \$20.08 in 2028.		-GPM% grows to 46.4% and EPS to \$17.38 in 2028.	

Risk Reward – MKS Inc. (MKS.O)

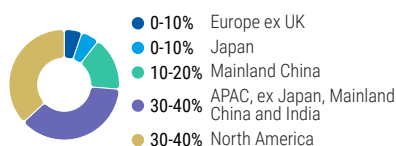
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
GAAP Revenue (\$, mm)	3,930	4,870	6,001	6,597
MW Gross Margin (%)	46.8	47.1	47.7	47.9
MW EPS (\$)	7.88	11.95	17.62	20.08
Inventory (\$, mm)	651	940	1,144	1,200
DOI (\$, mm)	158.5	154.8	152.9	145.9

INVESTMENT DRIVERS

- WFE and semiconductor subcomponent cycle.
- MKS' market share in semi subcomponents (outsized exposure to NAND).
- Advanced Electronics and PCB & Substrate cycle.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	4/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger than expected WFE spend
- Market share gains in semis on the back of NAND growth
- PC, smartphone, and server strength
- Debt paydowns progress quicker than anticipated

RISKS TO DOWNSIDE

- WFE growth slows and NAND makers continue to be disciplined on capex
- PC, smartphone, and server extended weakness
- Debt paydowns slow due to slow revenue growth and higher rates

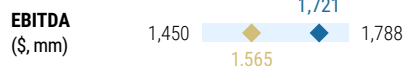
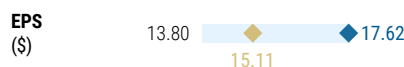
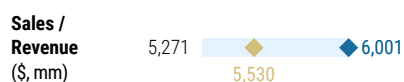
OWNERSHIP POSITIONING

Inst. Owners, % Active	63.1%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	29.5%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Nova Ltd (NVMI.O)

Metrology/Inspection intensity beneficiary at a premium valuation

PRICE TARGET \$540.00

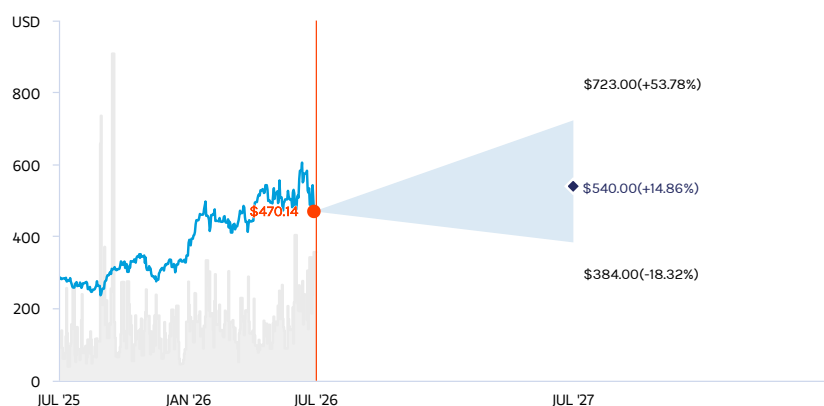
We use a 33x PE multiple on the company's CY28 EPS of \$15.39 (+ net cash), which implies a premium to its through-cycle average of 28x, commensurate with the multiple appreciation across the group.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

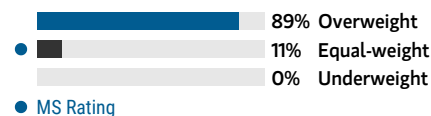
EQUAL-WEIGHT THESIS

■ NOVA has increased their share of WFE from an average of 32bp during 2018-2020 to 52bp during 2023-2025 driven by metrology/inspection intensity increasing in foundry logic and DRAM as well as NOVA's customer diversification and inorganic growth execution.

■ We expect the pace of share gains to moderate in 2026-27, but NOVA to gain share nonetheless driven by reacceleration of leading edge logic growth and continued share gains in DRAM.

■ NOVA trades at a premium multiple vs peers on our above-consensus estimates, which keeps us EW the name. We would be inclined to turn more positive if we get indication that NOVA can achieve our bull case EPS of \$15.50 in 2027 but stay sidelined for now.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*
Technology Diffusion: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$723.00

~39x CY28 non-GAAP EPS of \$17.72 + net cash

We reflect stronger than expected WFE market growth for 2026-2027 as well as market share gain and a multiple closer to early cycle.

- Revenue growth of 23.8% in CY28
- GM expands to 61.1% and EPS to \$17.72 in CY28

BASE CASE

\$540.00

~33x CY28 non-GAAP EPS of \$15.39 + net cash

Multiple is a premium to its through-cycle-average of 28x.

- Revenue growth of 12.6% in CY28
- CY28 GM of 60.1%
- EPS of \$15.39 in CY28

BEAR CASE

\$384.00

~26x CY28 non-GAAP EPS of \$13.53 + net cash

We reflect weaker than expected WFE market growth for 2026-2027 as well as market share loss and a multiple closer to late cycle.

- Revenue increases by 5.3% in CY28
- GM declines to 58.6% and EPS comes in at \$13.53 in CY28

Risk Reward – Nova Ltd (NVMI.O)

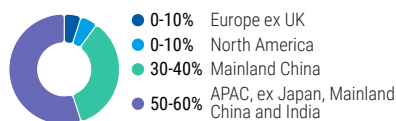
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
GAAP Revenue (\$, mm)	880.6	1,063.5	1,377.3	1,550.6
MW Gross Margin (%)	59.4	59.3	59.8	60.1
MW EPS (\$)	8.67	10.18	13.44	15.39
Inventory (\$, mm)	183.7	212.6	252.9	281.3
DOI	176.1	170.0	158.7	158.2

INVESTMENT DRIVERS

- Metrology/Inspection intensity in leading logic and memory
- Leading logic and memory capacity expansion
- Market share vs KLA & ONTO

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

RISKS TO PT/RATING

RISKS TO UPSIDE

- Metrology/inspection intensity increases in leading logic and memory
- Leading logic capacity expansion tracks ahead of expectation

RISKS TO DOWNSIDE

- Market share loss to KLA & ONTO
- Cyclical decline in memory leads to slower capex

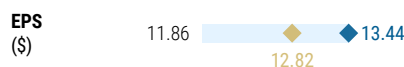
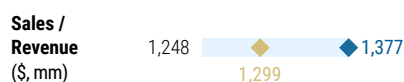
OWNERSHIP POSITIONING

Inst. Owners, % Active	79.3%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	29.5%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

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The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Shane Brett; Nicole Kozhukhov; Joseph Moore; Ella Tulchinsky; Mason Wayne.

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Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Applied Materials Inc., Camtek, MKS Inc., Nova Ltd.

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Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Applied Materials Inc., MKS Inc..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Applied Materials Inc., Camtek, KLA Corp, Lam Research Corp, MKS Inc., Nova Ltd, ONTO Innovation Inc, Teradyne Inc.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Applied Materials Inc., MKS Inc..

Morgan Stanley & Co. LLC makes a market in the securities of MKS Inc., ONTO Innovation Inc.

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Global Stock Ratings Distribution

(as of June 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Coverage Universe			Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/02/2026)
Shane Brett		
Applied Materials Inc. (AMAT.O)	E (05/18/2026)	\$603.04
Camtek (CAMT.O)	E (12/01/2025)	\$142.50
KLA Corp (KLAC.O)	O (01/15/2026)	\$235.55
Lam Research Corp (LRCX.O)	O (05/18/2026)	\$351.41
MKS Inc. (MKS.I.O)	O (08/04/2024)	\$365.56
Nova Ltd (NVMI.O)	E (12/01/2025)	\$470.14
ONTO Innovation Inc (ONTO.N)	O (06/14/2026)	\$307.58
Teradyne Inc (TER.O)	E (07/30/2025)	\$369.09

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